



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
UNIT TEST
13-02-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS]

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

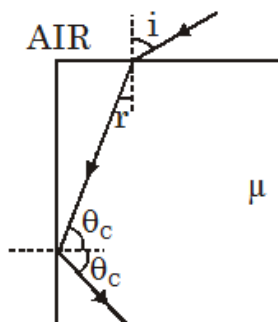
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. A particle is released from rest from a certain height $H = 80\text{m}$. Due to the wind the particle gathers the horizontal velocity component $v_x = ay$ where $a = 3/8 \text{ s}^{-1}$ and y is the vertical displacement of the particle from the point of release. Find the horizontal displacement of the particle when it strikes the ground. (Take : $g = 10 \text{ m/s}^2$)
 (A) 50 m (B) 40 m (C) 30 m (D) None of these
 2. The displacement of particle executing SHM is given by $x = A \sin (\omega t + \theta)$, where θ -initial phase of motion, A -Amplitude of motion and T -Time period. The magnitude of average velocity during motion of particle between $t = 0$ to $t = T/4$ is :
 (A) $2 \frac{A\omega}{\pi}$ (B) $2 \frac{A\omega}{\pi} |\cos\theta|$
 (C) $2 \frac{A\omega}{\pi} |\cos\theta - \sin\theta|$ (D) $2 \frac{A\omega}{\pi} |\cos\theta + \sin\theta|$
-

Space for Rough Work

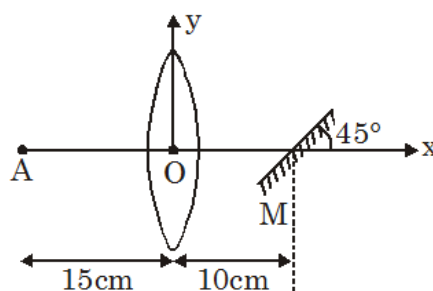
3. Two particles A and B are performing SHM along x and y-axis respectively with equal amplitude and frequency of 2 cm and 1 Hz respectively. Equilibrium positions of the particles A and B are at the coordinates (3, 0) cm and (0, 4) cm respectively. At $t = 0$, B is at its equilibrium position and moving towards the origin, while A is nearest to the origin and moving away from the origin. If smallest separation between A and B is $d_{\min}$ and maximum separation is $d_{\max}$ then $d_{\max} - d_{\min} = \underline{\hspace{2cm}}$?
- (A) 4 cm (B) 5 cm (C) 2 cm (D) 1.5 cm
4. In the figure shown, for an angle of incidence i at the top surface, what is the minimum refractive index needed for total internal reflection at the vertical face?



- (A) $\sqrt{1 - \sin^2 i}$ (B) $\sqrt{1 + \sin^2 i}$
- (C) $\sqrt{1 - \cos^2 i}$ (D) $\sqrt{1 + \cos^2 i}$

Space for Rough Work

5. A point A is kept on the axis of a converging lens of focal length 10 cm as shown in the figure. A plane mirror M is placed inclined at 45° to the lens axis at a distance of 10 cm to the right of the lens. The co-ordinates of the image of A formed by this combination are (take the centre of the lens as the origin and the lens axis as the axis of x). (Consider only two optical phenomenon)



- (A) (-10 cm, 20 cm) (B) (-10 cm, -20 cm)
 (C) (10 cm, -20 cm) (D) (10 cm, 20 cm)
6. A projectile is required to hit a target whose co-ordinates relative to horizontal and vertical axes through the point of projection are (α, β) . If the gun velocity is $\sqrt{2g\alpha}$, then the condition when it is impossible to hit the target, is
- (A) $3\alpha > 4\beta$ (B) $4\beta > 3\alpha$ (C) $3\beta > 2\alpha$ (D) $4\alpha > 3\beta$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

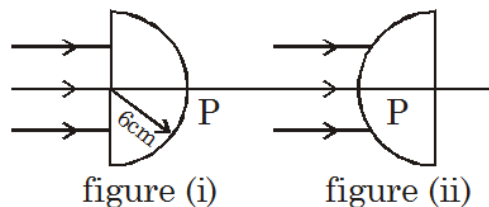
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

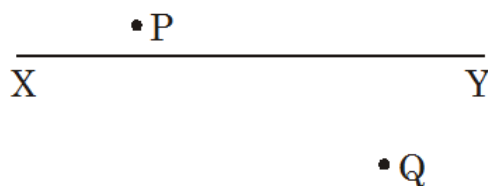
7. A parallel beam of light is incident normally on the flat surface of a hemisphere of radius 6cm and refractive index 1.5, placed in air as shown in figure (i). Assume paraxial ray approximation. Choose the correct option(s) :-



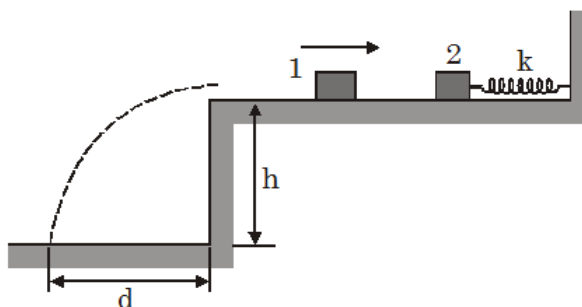
- (A) The rays are focussed at 12 cm from the point P to the right, in the situation as shown in figure (i)
- (B) The rays are focussed at 16 cm from the point P to the right, in the situation as shown in figure (i)
- (C) If the rays are incident at the curved surface (figure (ii)) then these are focused at distance 18 cm from point P to the right
- (D) If the rays are incident at the curved surface (figure (ii)) then these are focused at distance 14 cm from point P to the right

Space for Rough Work

8. Two points P and Q lie on either side of an axis XY as shown. It is desired to produce an image at Q for a real object placed at P using a spherical mirror, with XY as the optic axis. The mirror must be



- (A) converging and positioned to the left of P (B) diverging and positioned to the left of P
(C) converging and positioned to the right of Q (D) diverging and positioned to the right of Q
9. Figure shows block 1 of mass 0.5 kg sliding to the right over a frictionless elevated surface at a speed of 12 m/s. The block undergoes an elastic collision with stationary block 2, which is attached to a spring of spring constant 100 N/m (Assume that spring doesn't affect collision). After the collision, block 2 oscillates in SHM with a time period of 0.628 s and block 1 slides to the opposite end of the elevated surface landing at a distance d from the base of that surface after falling through a height $h = 4.90$ m. Then which of the following is (are) **CORRECT**: (take $g = 9.8 \text{ m/s}^2$ and $\pi = 3.14$)



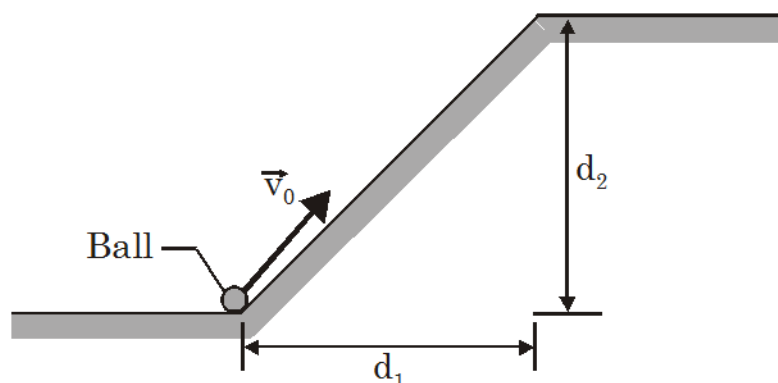
- (A) The value of $d = 2$ m
(B) Spring potential energy is < 35 J
(C) Maximum kinetic energy of block 2 is 32 J
(D) Speed of block 1 when it hits the ground is 4 m/s

Space for Rough Work

10. Two boats are at sea and moving continuously in straight lines. At 11 AM boat 1 was 40 km North of a small island and was moving at 15 km/h towards east. The boat 2 at 7 AM of the same day was 100 km away east of the same island and moved at a speed of 15 km/h towards the South. Which of the following is (are) **CORRECT**?
- (A) The minimum separation between the boats is $100\sqrt{2}$ km
 - (B) The minimum separation between the boats is less than $100\sqrt{2}$ km
 - (C) The boats will be found at minimum separation at 9 AM
 - (D) The boats will be found at minimum separation at 11 AM

Space for Rough Work

11. In the figure shown, a ball is launched with a velocity of magnitude 10 m/s at an angle 53° with the horizontal. The launch point is at the base of a ramp of horizontal length d_1 and height d_2 . A plateau (flat surface) is located at the top of the ramp. Which of the following option(s) is(are) **CORRECT**? (Take : $g = 10 \text{ m/s}^2$)



- (A) If $d_1 = 12 \text{ m}$ and $d_2 = 5 \text{ m}$, the ball lands on ramp.
- (B) If $d_1 = 6 \text{ m}$ and $d_2 = 5 \text{ m}$, ball lands on plateau.
- (C) If $d_1 = 6 \text{ m}$ and $d_2 = 3 \text{ m}$, the displacement of the ball from the point of projection to the point where it hits the surface is $3\sqrt{5} \text{ m}$
- (D) If $d_1 = 6 \text{ m}$ and $d_2 = 1 \text{ m}$, displacement of the particle is $> 3\sqrt{5} \text{ m}$
12. A particle moves along the x-axis according to the law $x = a \sin^2 \left(\omega t - \frac{\pi}{4} \right)$
- (A) The amplitude of the oscillation is a
- (B) The time period of oscillations is $\frac{\pi}{\omega}$
- (C) The velocity (v) of the particle is given by $v^2 = 4\omega^2 x (a - x)$
- (D) The velocity (v) of the particle is given by $v^2 = 2\omega^2 x (a - x)$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

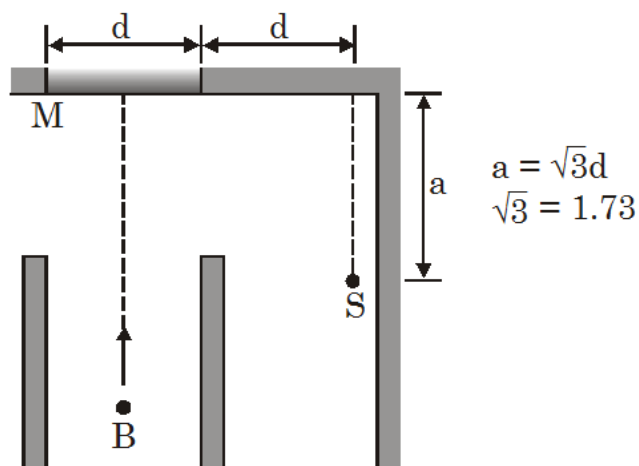
Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. When the plane surface of plano-convex lens is silvered, the lens behaves like a concave mirror of focal length 30 cm, but when the convex surface is silvered, it behaves like a concave mirror of focal length 10 cm. The refractive index of material of lens is $\frac{P}{2}$. What is the value of P?
 2. A bird flies with a uniform speed of 10 km/h and a car moves with a uniform speed of 8 km /h. Both start from B towards A (BA = 40 km) at the same instant. The bird having reached A, flies back to meet the approaching car. As soon as it reaches the car, it flies back to A. Assume that bird immediately gains the same speed while turning back. The distance of the car from point A is $\frac{16}{\alpha}$ when the fly reaches point A for the 3rd time. The value of α is :
-

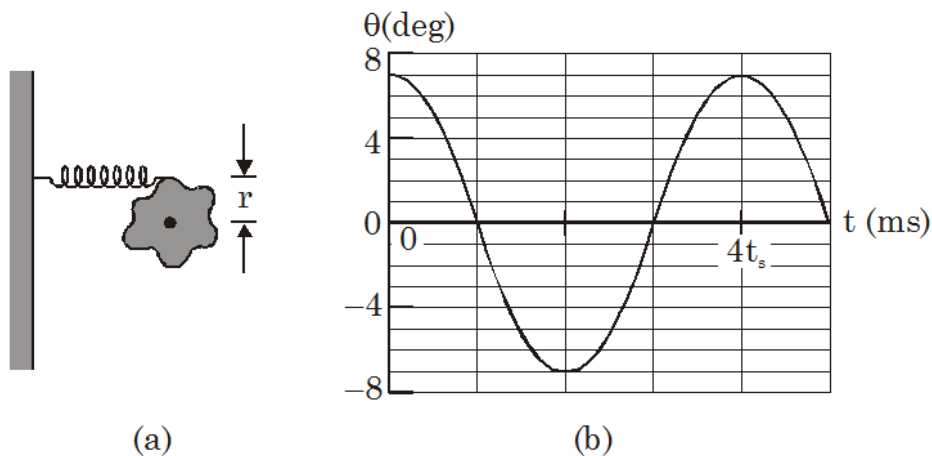
Space for Rough Work

3. A ray of light passes through a prism in principle plane. The angle of prism is α . Angle of deviation and angle of incidence both are equal to 2α . If μ is the refractive index of the prism then it can be found that $\cos^2 \alpha = \frac{\mu^n - 1}{2x}$ where n is a positive integer. The value of x is:
4. Figure shows an overhead view of a corridor with a plane mirror M mounted at one end. A burglar B sneaks along the corridor directly towards the centre of the mirror. If $d = 4\text{m}$, how far (in meter) from the mirror is the burglar when he will be first seen in the mirror by the security guard S ?

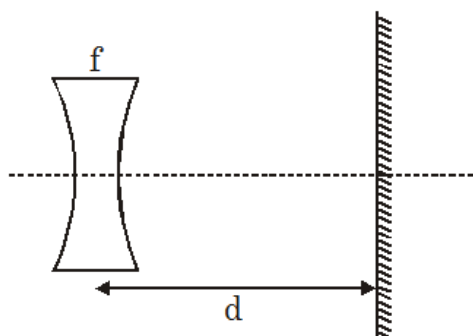


Space for Rough Work

5. A metal plate is mounted on an axle through its centre of mass. A spring with spring constant $k = 2000$ N/m connects a wall with a point on the rim a distance $r = 2.5$ cm from the centre of mass. Initially the spring is at its rest length. If the plate is rotated by 7° and released, it rotates about the axle in SHM with its angular position given by the graph shown. The horizontal axis scale is set by $t_s = 31.4$ ms. If the moment of inertia of the plate about the centre of mass is $x \times 10^{-6}$ kg m². Then the value of x is:



6. A diverging lens of focal length 10 cm is placed 10 cm in front of a plane mirror as shown in the figure. Light from a very far away source falls on the lens. The final image is at a distance "V" cm from the mirror. Write the value of V (positive) :



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
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- For each question, choose the option corresponding to the correct answer.
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Full Marks : +3 If **ONLY** the correct option is chosen.

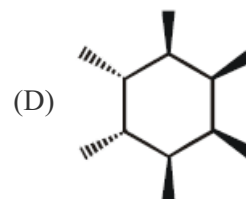
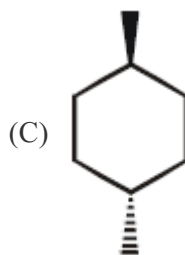
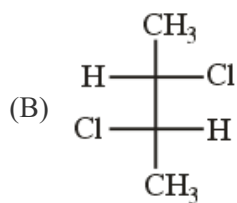
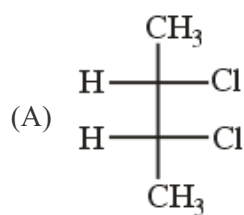
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. On mixing 15.0 ml of ethyl alcohol of density 0.792 g ml^{-1} with 15 ml of pure water at 4°C , the resulting solution is found to have a density of 0.924 g ml^{-1} . The percentage contraction in volume is :
 (A) 8 % (B) 2 % (C) 3 % (D) 4 %
 2. Which of the following is intrinsic semiconductor?
 (A) Si (B) Si doped with P
 (C) GaAs (D) $\text{Fe}_{0.93}\text{O}_{1.00}$
-

Space for Rough Work

3. Which of the following compound is optically active with C_2 -axis of symmetry ?

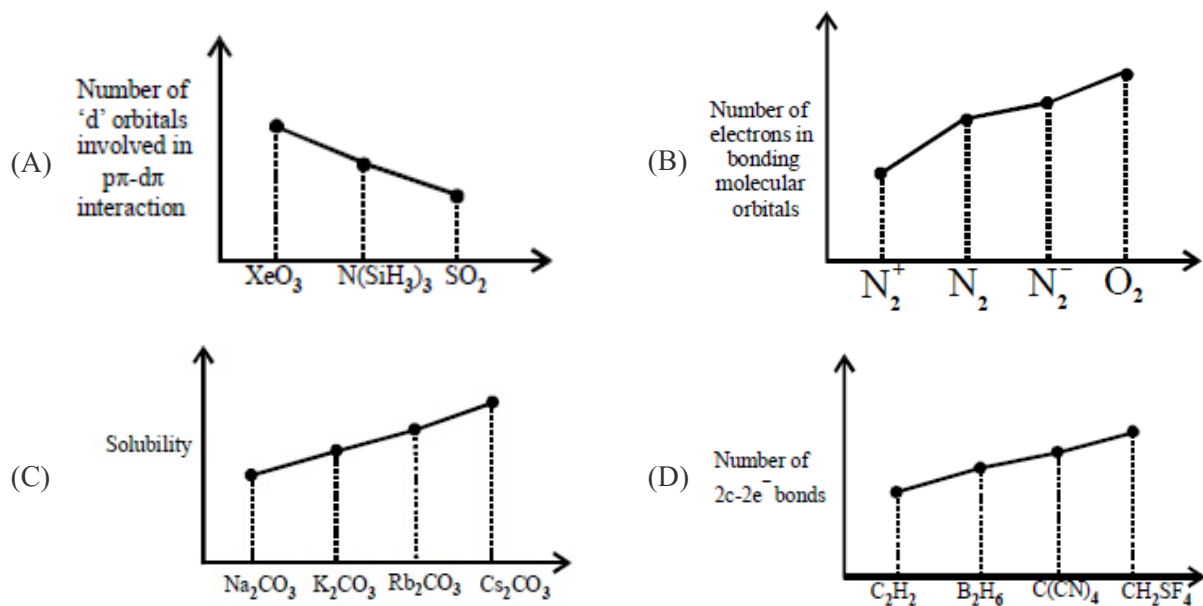


4. Select correct statement(s) among the following

- (A) Polar protic solvent favours S_N2 mechanism more than polar aprotic solvent generally
- (B) Nucleophilic order of halides in polar protic solvent is $I^- > Br^- > Cl^- > F^-$
- (C) Always perfect racemisation take place in S_N1 mechanism
- (D) Inversion of configuration in S_N2 mechanism always produces R to S configuration or S to R configuration

Space for Rough Work

5. Which of the following graph correctly depicts the property mentioned against the indicated species.



6. The specie that show maximum number of stereo Isomers among the following.

- (A) $[\text{Ni}(\text{H}_2\text{O})_2(\text{en})_2]^{2+}$ (B) $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$
 (C) $[\text{Pt}(\text{py})_2(\text{NH}_3)_2\text{Cl}_2]^{2+}$ (D) $[\text{CoBr}_2\text{Cl}_2(\text{NO}_2)(\text{SCN})]^{3-}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
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- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. A metal 'M' crystallizes in two crystal systems fcc and hcp. If d -density, z - number of atoms per unit cell and a, b, c are edge lengths of the unit cell along the crystallographic axes then. Select the correct options :

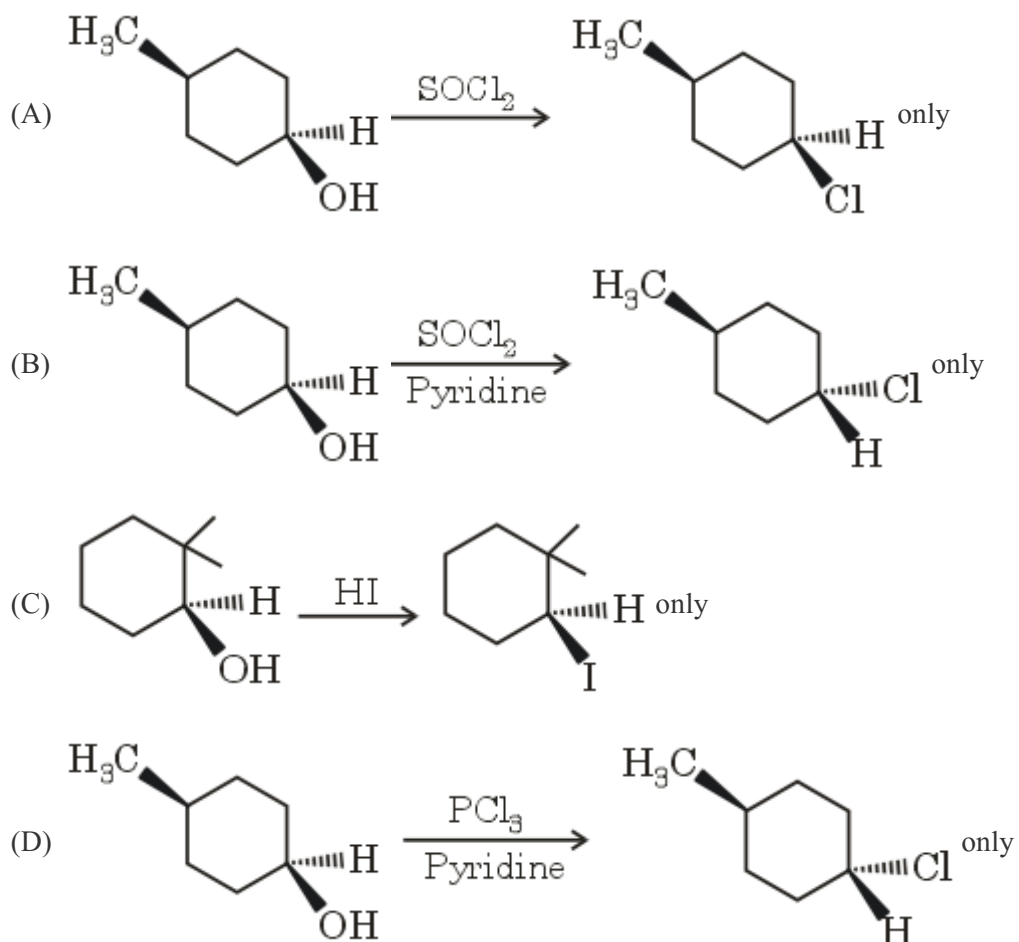
(A) $\frac{d_{fcc}}{d_{hcp}} \neq 1$ (B) $\frac{a_{fcc}}{c_{hcp}} = \frac{\sqrt{3}}{2}$ (C) $\frac{d_{fcc}}{d_{hcp}} = 1$ (D) $\frac{z_{fcc}}{z_{hcp}} = \frac{2}{3}$

8. The correct information(s) about CaF_2 crystal (Fluorite structure) is/are.

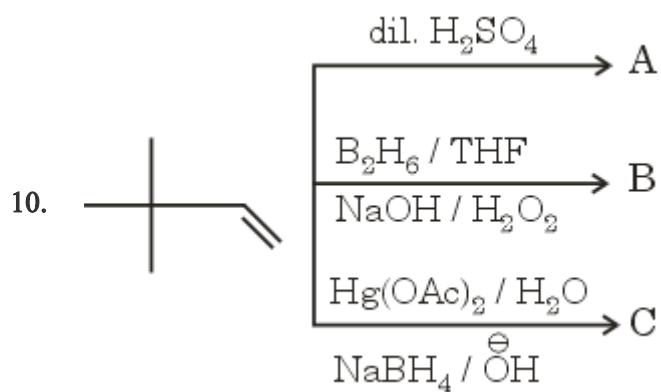
- (A) C.N. of Ca^{2+} ion is 8
 (B) C.N. of F^- ion is 8
 (C) F^- ions occupy all the tetrahedral voids formed by CCP arrangement of Ca^{2+} ions
 (D) Ca^{2+} ions occupy all the cubic voids formed by simple cubic arrangement of F^- ions

Space for Rough Work

9. Which of the following reaction is showing correct major product :



Space for Rough Work

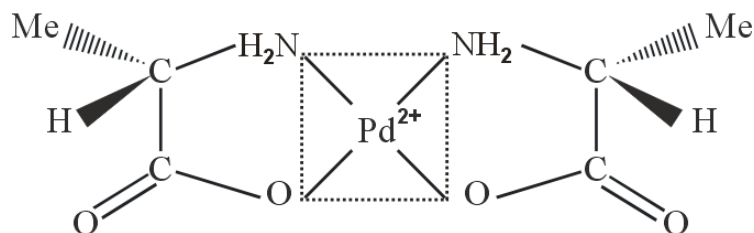


Select correct statement from the following about major product A, B and C :

- (A) A is 3° alcohol and show optical isomersim
- (B) B is 1° alcohol and does not show optical isomersim
- (C) C is 2° alcohol and shows optical isomersim
- (D) A and B are positional isomer of each other

Space for Rough Work

11. One of the isomer of $[\text{Pd}(\text{NH}_2\text{--CH}(\text{CH}_3)\text{--CO}_2^-)_2]$ is given as



The correct statement(s) regarding the above compound is/are.

- (A) Total stereoisomers will be 6
 - (B) The ligand $(\text{NH}_2\text{--CH}(\text{CH}_3)\text{--COO}^-)$ is Ambidentate ligand
 - (C) The molecule is planar since Pd^{2+} shows square planar geometry around it
 - (D) Exactly two Enantiomeric pairs will be formed
12. The compound AB_3 has more dipole moment than AC_3 . The compound AC_3 on action of superheated steam under high pressure yield an acidic oxide via $\text{S}_{\text{N}}1$ mechanism. AB_3 has tendency to unsymmetrically cleave bridge bond of B_2H_6 . Which is/are correct in view of above information.
- (A) The compound "BC" can be produced by hydrolysis of C_2 .
 - (B) The compound BA_3 has less B–A bond length compared to C–A bond length in CA_3 .
 - (C) AC_3 show back bonding.
 - (D) A can form neutral oxide

Space for Rough Work

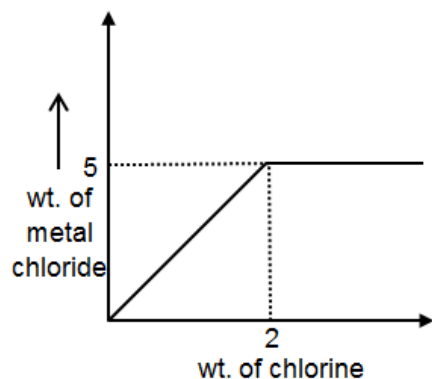
SECTION-II : (Maximum Marks: 24)

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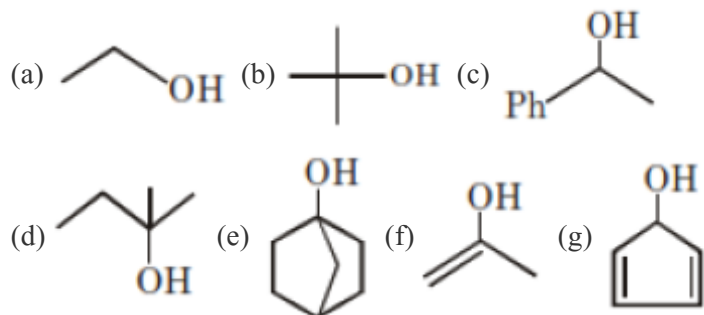
1. A compound has cubical unit cell in which 'A' atoms are found at six corners and 'B' atoms are present at remaining corners and at face centres which are not opposite to each other and 'C' atoms are present at body centres and remaining face centres. Calculate density of the compound in $\text{amu}/\text{\AA}^3$.
Given : A = 20 amu, B = 60 amu, C = 80 amu and edge length = $2\text{\AA}$.
2. A metal chloride of empirical formula MCl is formed (where M is metal). In an experiment weight of metal chloride has following variation according to weight of chlorine.



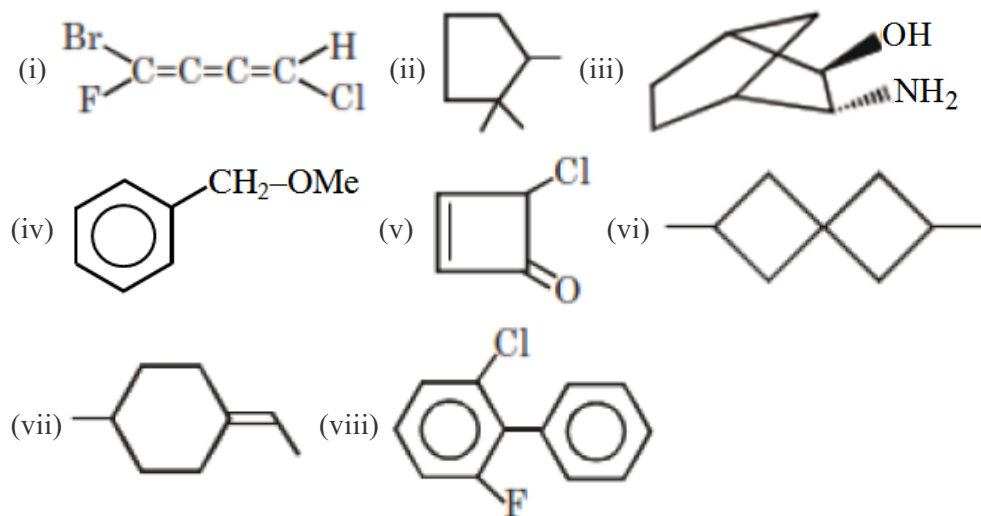
Then atomic weight of metal (M) will be :

Space for Rough Work

3. How many of the following compound will form immediate turbidity with (HCl / ZnCl₂):



4. Number of molecules show optical isomerism :



Space for Rough Work

5. Which of the following compounds/species are incorrectly matched against the indicated names.
- (a) $[\text{PdI}_2(\text{ONO})_2(\text{H}_2\text{O})_2] \rightarrow$ Diaquadiiodidodinitrito-Npalladium(IV)
- (b) $\text{Fe}(\pi\text{-C}_5\text{H}_5)_2 \rightarrow$ Bis(η^1 -cyclopentadienyl)iron(II)
- (c) $\text{H}_2[\text{PtCl}_6] \rightarrow$ Hexachloridoplatinate(IV) hydride
- (d) $[\text{Al}(\text{OH})(\text{H}_2\text{O})_5]\text{SO}_4 \rightarrow$ Pentaquahydroxidoaluminium(III) sulphate
- (e) $\text{K}_2[\text{OsCl}_5\text{N}] \rightarrow$ Potassium nitridopentachloridoosmate(VI)
- (f) $\text{Cr}(\text{C}_6\text{H}_6)_2 \rightarrow$ Bis(η^6 -benzene)chromium(0)
6. On the basis of the following observations made with aqueous solutions of given complexes, assigning secondary valency of each metal, write the sum of secondary valencies of all given complexes as your answer.

Formula	Moles of AgCl precipitated per mole of compound with excess AgNO_3 solution
(i) $\text{PdCl}_2.4\text{NH}_3$	2 moles
(ii) $\text{NiCl}_2.6\text{NH}_3$	2 moles
(iii) $\text{CoCl}_3.4\text{NH}_3$	1 mole
(iv) $\text{PtCl}_2.2\text{NH}_3$	Zero mole

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Consider a hyperbola $x^2 - y^2 - 2x + 4y - 7 = 0$. If its pair of asymptotes meet two tangents each of slope 2 at P, Q and R, S (in order) then choose the **INCORRECT** statement :
 (A) $PQ = \frac{4\sqrt{5}}{\sqrt{3}}$ (B) Latus rectum of hyperbola is 4
 (C) area of quadrilateral PQRS is 12 sq.units (D) area of quadrilateral PQRS is 16 sq.units
2. Consider $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{3x^2 + ax + b}{x^2 + 1}$ where $a, b \in \mathbb{R}$. If the range of this function is $[-4, 3)$ then $a + b$ is equal to
 (A) 0 (B) -4 (C) -1 (D) 3
3. Given that $f(x)$ is a function defined on $\mathbb{R}$, satisfying $f(1) = 1$ and for any $x \in \mathbb{R}$, $f(x + 5) \geq f(x) + 5$ and $f(x + 1) \leq f(x) + 1$. If $g(x) = f(x) + 5 - x$, then $g(2021)$ is equal to -
 (A) 3 (B) 4 (C) 5 (D) 6

Space for Rough Work

4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $f(x)$ be a polynomial with real coefficient such that $(x^2 + x + 1) f(x - 1) = (x^2 - x + 1) f(x) \forall x \in \mathbb{R}$ and $f(0) = 1$, then which of the following is false -
- (A) $f(x)$ is many one function (B) $f(x)$ has no real roots
- (C) range of $f(x)$ is $\left[\frac{1}{2}, \infty\right)$ (D) $f(x)$ is neither even nor odd
5. If $[x]^3 + [y]^3 + [z]^3 - 3[x][y][z] = 4$; where $x, y, z > 0$ (where $[.]$ denotes GIF); then minimum possible value of $(x + y + z)$ is
- (A) 2 (B) 3 (C) 4 (D) 6
6. Solution set of equation is $\left|1 - \log_{\frac{1}{6}} x\right| + |\log_2 x| + 2 = \left|3 - \log_{\frac{1}{6}} x + \log_{\frac{1}{2}} x\right|$, $a, b \in \mathbb{N}$, then the value of $(a + b)$ is
- (A) 5 (B) 6 (C) 7 (D) 8

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. Tangent is drawn at any point (x_1, y_1) other than the vertex, on parabola $y^2 = 4ax$. If tangents are drawn from any point on this tangent to the circle $x^2 + y^2 = a^2$ such that all the chords of contact pass through a fixed point (x_2, y_2) then

(A) x_1, a, x_2 are in GP

(B) $\frac{y_1}{2}, a, y_2$ are in GP

(C) $-4, \frac{y_1}{y_2}, \frac{x_1}{x_2}$ are in GP

(D) $x_1x_2 + y_1y_2 = a^2$

Space for Rough Work

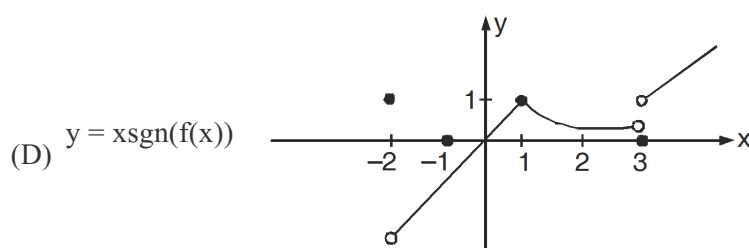
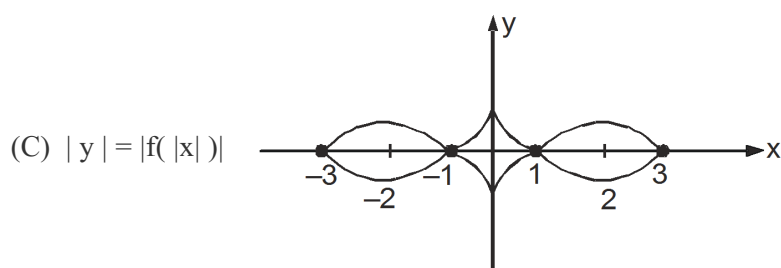
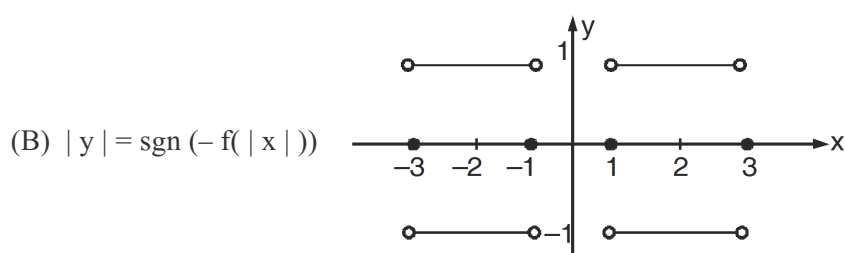
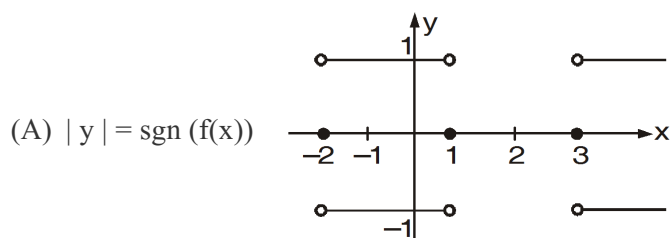
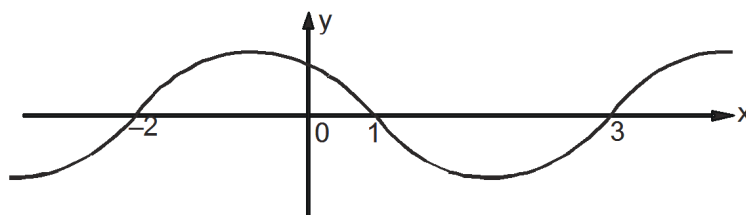
8. A straight line $y = m(x - a)$, $m \neq 0$ meets a parabola $y^2 = 4ax$ at two points A and B. Then which of the following option(s) is/are correct
- (A) Length $AB = 4a$ when $m = 1$
- (B) Length $AB = 8a$ when $m = 1$
- (C) Angle between normals drawn at A and B is 45° when $m = 1$
- (D) Angle between normals drawn at A and B is 90° when $m = 2$
9. If tangents PQ and PR are drawn from a point on the circle $x^2 + y^2 = 25$ to the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$, ($b < 4$), so that the fourth vertex S of the parallelogram PQSR lies on the circumcircle of triangle PQR, then which of the following is/are true
- (A) $e = \frac{\sqrt{7}}{4}$
- (B) $e = \frac{\sqrt{5}}{4}$
- (C) $b = 3$
- (D) $b = \sqrt{11}$

Space for Rough Work

10. Let ordered pair (x, y) satisfy the equation $49^{x^2+y} + 49^{y^2+x} = \frac{2}{\sqrt{7}}$ then which of the following is/are correct ?
- (A) Only 1 ordered pair of (x, y) is possible
- (B) More than 1 ordered pair of (x, y) are possible
- (C) In one of the ordered pair $|x|+|y| = 1$
- (D) In one of the ordered pair $x - y = 0$
11. The values of x satisfying the equation $x^9 + \frac{9}{8}x^6 + \frac{27}{64}x^3 - x + \frac{219}{512} = 0$ is -
- (A) $\frac{-1 + \sqrt{13}}{4}$
- (B) $-\frac{1}{2}$
- (C) $\frac{-1 - \sqrt{13}}{4}$
- (D) $\frac{1}{2}$

Space for Rough Work

12. The graph of the function $y = f(x)$ is shown in the figure. Then which one of the following graph(s) of corresponding curve is/are correct ?



(Here $\text{sgn}(\cdot)$ denotes signum function)

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. Let $A = \{1, 2, 3, 4\}$, $B = \{1, 2, 3, 4\}$. The total number of functions $f : A \rightarrow B$ such that $f(x) \neq x \forall x \in A$ and range of $f(x)$ contains exactly three elements, is
 2. Let T_n be the n^{th} term of a series for $n \geq 1$ and $T_1 = 1$, $T_{n+1} = 2T_n + 4^n$, $n \geq 1$. If S_n = Sum of first n terms of the series and $S_n = \frac{a \cdot 4^n + b \cdot 2^n + 1}{3}$ where $a, b \in I$, then the value of $\left| \frac{b}{a} \right|$ is
 3. Let a, b and c are distinct positive numbers, none of them is equal to unity such that $\log_b a \cdot \log_c a + \log_a b \cdot \log_c b + \log_a c \cdot \log_b c - \log_b a \sqrt{a} \log_{\sqrt{c}} b^{1/3} \log_a c^3 = 0$, then the value of abc is -
-

Space for Rough Work

4. If the fundamental period of the function $f(x) = \sin x + \tan \frac{x}{2} + \sin \frac{x}{2^2} + \tan \frac{x}{2^3} + \dots + \sin \frac{x}{2^7} + \tan \frac{x}{2^8}$ is $k\pi$, then k is equal to -
5. Two perpendicular normals are drawn at the point P and Q of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, intersecting each other on the major axis at a point A (other than origin). If the area of $\triangle APQ$ is λ , then the value of λ equals
6. Area enclosed within the curve $[y] = 2^{[x]}$, $x \in [0, 9)$ (where $[\cdot]$ denotes the greatest integer function) is equal to

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Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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